

STAT 1210

Chapter 1 Homework

Fall 2026

Question 1

A greenhouse researcher recorded the following information for six seedlings.

Plant ID	Species	Treatment	Number of leaves	Height (cm)
P101	Bean	Control	7	18.4
P102	Bean	Fertilizer	10	22.1
P103	Sunflower	Control	8	25.6
P104	Sunflower	Fertilizer	12	31.2
P105	Tomato	Control	9	20.8
P106	Tomato	Fertilizer	13	27.5

1. Who are the individuals in this dataset?
2. Which four variables were recorded about each individual?
3. What is the purpose of the Plant ID column? Why should it not be treated as a quantitative measurement?

Solution

1. The individuals are the six seedlings represented by the six rows of the table.
2. The four variables are species, treatment, number of leaves and height. Plant ID labels the individuals rather than measuring a characteristic of interest.
3. Plant ID allows each seedling to be identified. Although the labels contain numbers, their numerical values do not represent amounts. So arithmetic with them would have no useful meaning.

Question 2

A bird-monitoring project records the following information for each nest:

- the bird species;
 - the nesting stage, recorded as building, eggs, hatchlings, or fledged;
 - the number of eggs;
 - the average egg mass in grams; and
 - the nest identification code.
1. Classify each characteristic as categorical or quantitative.
 2. For each categorical variable, state whether it is nominal or ordinal.
 3. For each quantitative variable, state whether it is discrete or continuous.
 4. Why is the nest identification code different from an ordinary categorical variable of scientific interest?

Solution

1. Bird species and nesting stage are categorical. Number of eggs and average egg mass are quantitative. The nest code is an identifier.
2. Bird species is nominal because its categories have no natural order. Nesting stage is ordinal because the stages follow a meaningful biological order.
3. Number of eggs is discrete because it is a count. Average egg mass is continuous because it is measured and can, in principle, take any value in an interval.
4. The nest code is used to distinguish records. Its labels are not outcomes whose distribution would normally answer the scientific question.

Question 3

For each situation, select the most appropriate graph from the following list: **bar graph, pie chart, dotplot, histogram, or time plot**. Briefly justify each choice.

1. The blood types of 250 blood donors are recorded.
2. The germination times, in days, of 12 seeds are recorded.
3. The body temperature of one animal is measured every hour for 48 hours.
4. The shell lengths of 500 mussels are measured.
5. A laboratory reports the percentage of samples in four outcome categories that together account for every sample.

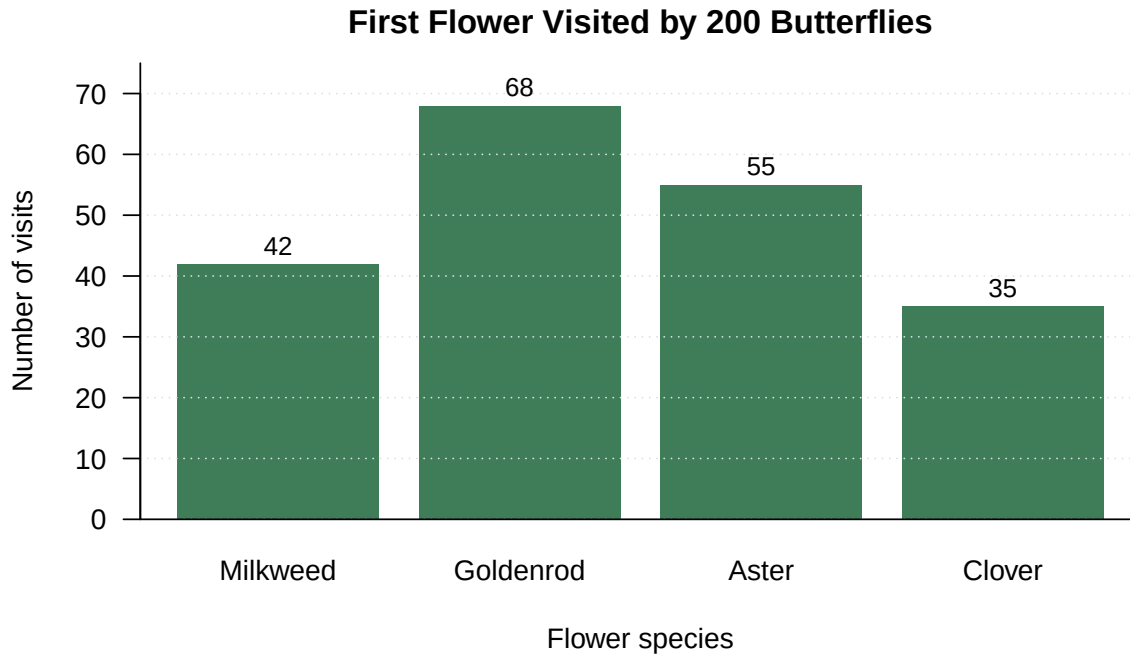
Solution

1. A bar graph is appropriate because blood type is categorical and the separated bars can compare category frequencies.
2. A dotplot is appropriate because germination time is quantitative and the sample is small enough to display every observation.
3. A time plot is appropriate because the order of the hourly measurements is essential for detecting trends or cycles.
4. A histogram is appropriate because shell length is quantitative and the sample is large.
5. A pie chart is permissible because the four percentages are parts of the same whole and sum to 100%. A bar graph would also be appropriate and would usually make comparisons easier.

Question 4

During a field survey researchers recorded the first flower species visited by each of 200 butterflies.

Flower species	Number of visits
Milkweed	42
Goldenrod	68
Aster	55
Clover	35



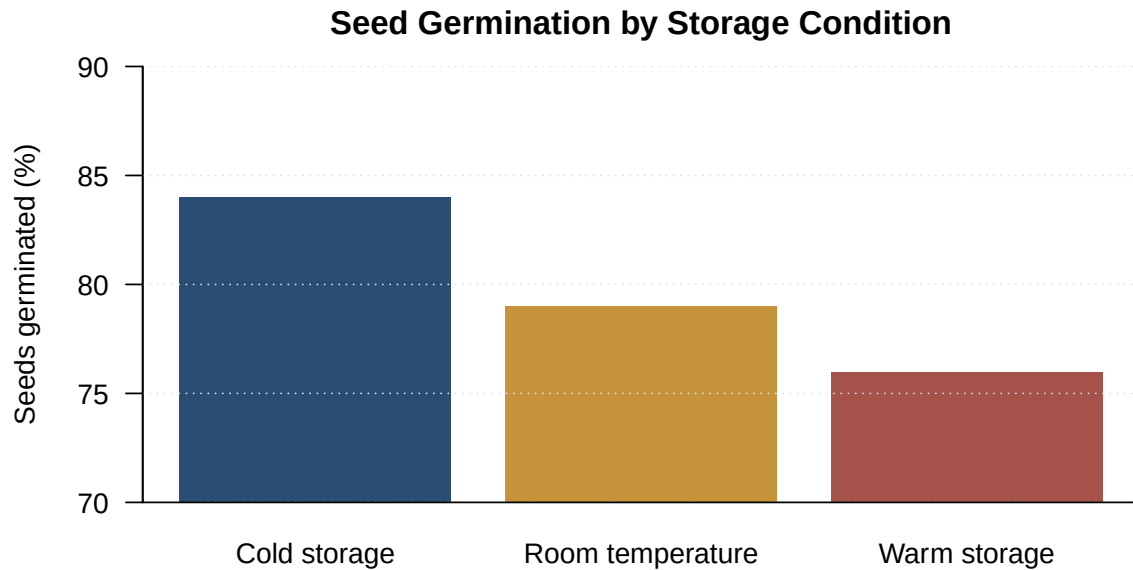
1. What percentage of visits occurred on each flower species?
2. Which category is the modal category and what does that mean in context?
3. Why are the bars separated rather than touching?
4. Would a pie chart be valid for these data? Why or why not?

Solution

1. The percentages are 21% for milkweed, 34% for goldenrod, 27.5% for aster and 17.5% for clover.
2. Goldenrod is the modal category because it has the largest frequency with 68 of the 200 first visits.
3. The bars are separated because flower species is categorical. The spaces emphasize that the categories are distinct rather than intervals on a numerical scale.
4. Yes. Every butterfly contributes to exactly one category so the categories are parts of one whole and their percentages sum to 100%. A bar graph still makes close comparisons easier.

Question 5

The graph below compares the percentage of seeds that germinated under three storage conditions. Each percentage is based on a separate group of seeds.



1. What feature of the graph could create a misleading impression?
2. What is the actual difference, in percentage points, between cold storage and warm storage?
3. How should the graph be corrected?
4. Can the number of germinated seeds in each group be determined from this graph? Explain.

Solution

1. The vertical axis begins at 70% rather than 0%. Because the visible bar lengths are severely shortened, the graph exaggerates the differences among the three rates.
2. The difference is $84\% - 76\% = 8$ percentage points.
3. A corrected bar graph should begin its vertical axis at 0%. The title and axis labels should remain clear, and the group sample sizes should be reported.
4. No. A percentage does not reveal a count unless the denominator is known. For example, 84% could represent 84 of 100 seeds or 420 of 500 seeds.

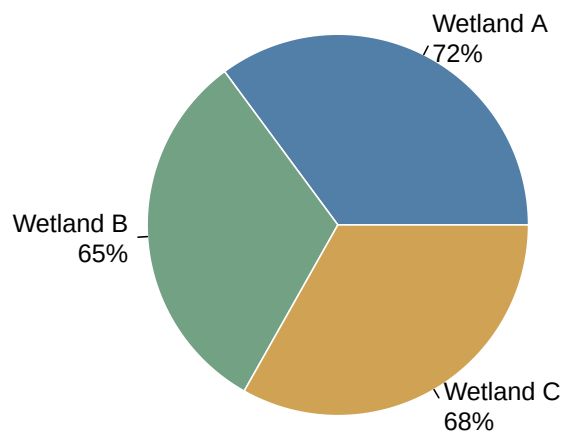
Question 6

A researcher recorded whether a frog species was detected during surveys at three wetlands.

Wetland	Surveys with a detection	Total surveys	Detection percentage
A	18	25	72%
B	130	200	65%
C	34	50	68%

The researcher placed the three percentages into one pie chart.

Proposed Display of Wetland Detection Percentages



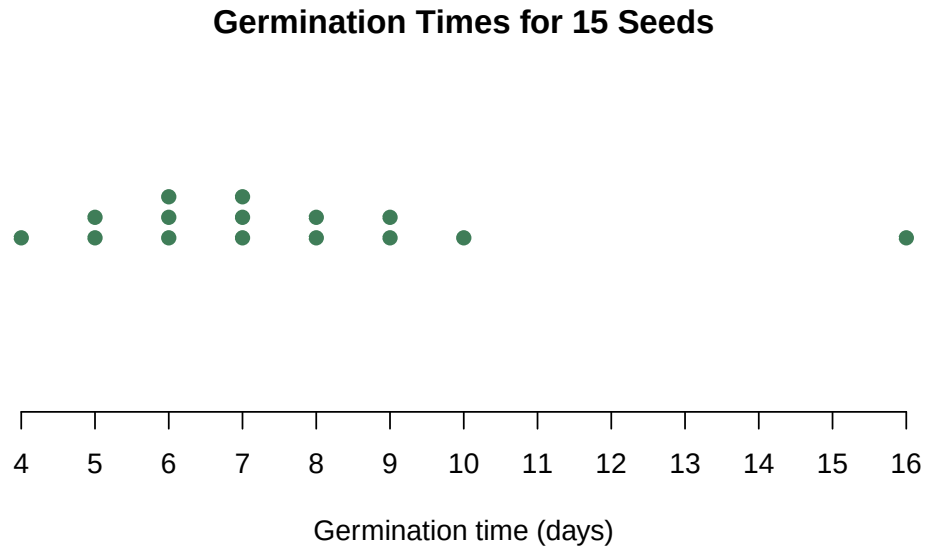
1. Why is this pie chart not a valid representation of the data?
2. What does the sum of the three displayed percentages reveal about the problem?
3. Which wetland has the highest detection percentage? Which wetland has the largest number of surveys with a detection?
4. What display would communicate the comparison more accurately?

Solution

1. The percentages use three different denominators and do not represent parts of one common whole. A pie chart incorrectly forces them to appear as shares of a single total.
2. The percentages sum to 205%, not 100%. This confirms that the values cannot be slices of one whole.
3. Wetland A has the highest detection percentage at 72%. Wetland B has the largest number of surveys with a detection, with 130, because it also has a much larger number of surveys.
4. A bar graph or dot plot of the three detection percentages would be appropriate. The total number of surveys for each wetland should appear in the labels, table or caption.

Question 7

The following dotplot shows the number of days required for 15 seeds to germinate.



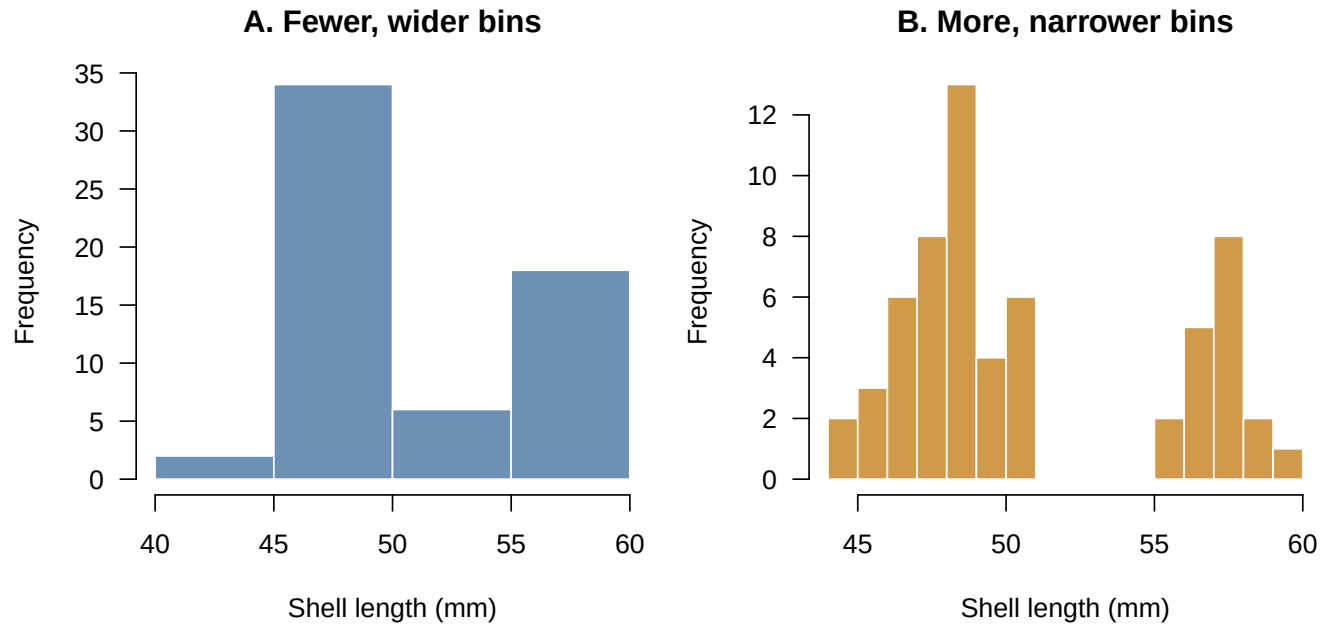
1. Is germination time categorical or quantitative? Is it discrete or continuous as recorded here?
2. Describe the distribution using **SOCS**: shape, outliers, centre and spread.
3. Should the observation at 16 days automatically be deleted? What should happen first?

Solution

1. Germination time is quantitative. Because it is recorded in whole days in this dataset, it is discrete as recorded, although time itself can be measured continuously.
2. The distribution is unimodal and right-skewed. A possible outlier occurs at 16 days. The centre is approximately 7 days and the observations range from 4 to 16 days. Most observations lie between 5 and 10 days.
3. No. The original record should first be checked for a data entry or measurement error. If the value is correct, it may represent meaningful biological variation and should not be removed without a defensible reason.

Question 8

Two histograms below display the same synthetic measurements of mussel shell length. Only the number of bins has changed.



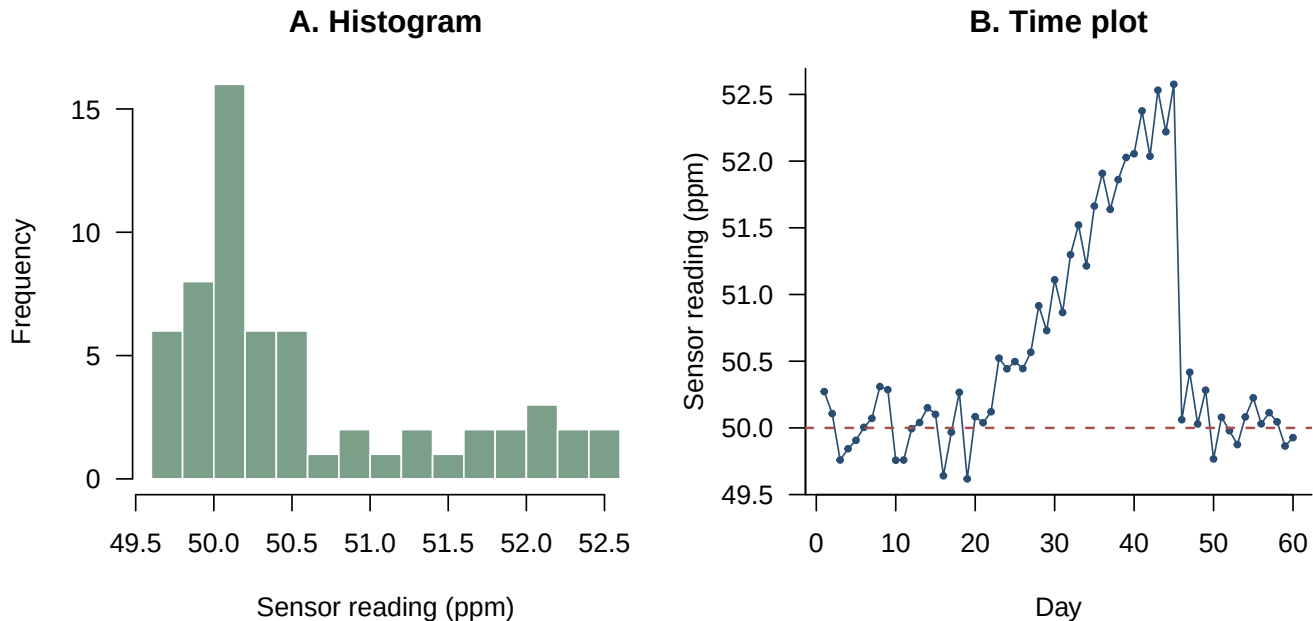
1. What characteristics of the distribution appear different between the two histograms?
2. Why can changing the bin width change the viewer's impression even though the data have not changed?
3. Which features of the data remain unchanged regardless of the bin width?
4. What is a responsible approach to choosing a histogram bin width?

Solution

1. The histogram with fewer bins gives a smoother and less detailed impression. The histogram with narrower bins more clearly suggests two clusters, one near 48 mm and another near 57 mm, but it also appears more irregular.
2. A histogram groups observations into intervals. Different interval boundaries and widths change which measurements are combined, so peaks, gaps, and apparent skewness can become more or less visible.
3. The sample size, individual measurements, overall minimum and maximum and underlying dataset do not change. Only the grouping used for the display changes.
4. Several reasonable bin widths should be examined. A useful width reveals important structure without creating excessive noise and the chosen width should not be selected merely to support a preferred conclusion.

Question 9

An air quality laboratory checks a gas sensor once per day using a certified reference gas with a concentration of 50 ppm. The sensor should produce readings close to 50 ppm, apart from small random measurement variation. The same 60 daily readings are displayed as a histogram and as a time plot. The data are illustrative but represent a realistic pattern of sensor drift followed by recalibration.



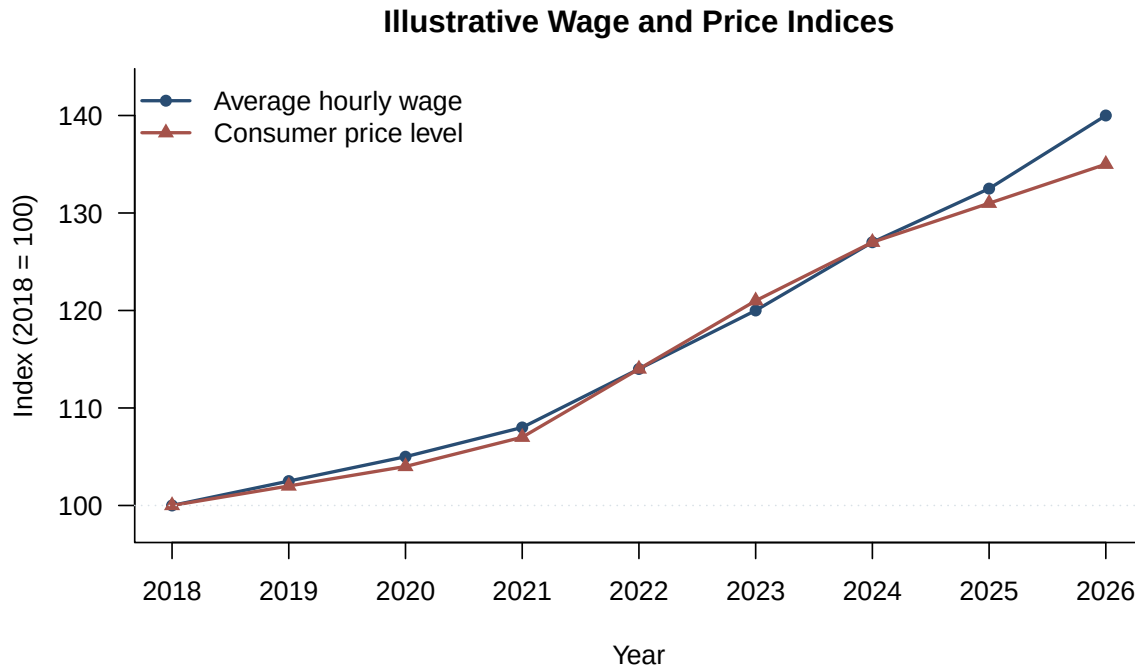
1. What information does the histogram communicate effectively?
2. What important pattern is hidden by the histogram but visible in the time plot?
3. Why is time order essential in this setting?
4. What likely measurement issue should the laboratory investigate and what might explain the sudden change around day 46?

Solution

1. The histogram summarizes the overall distribution of the readings, including their approximate range, clusters, centre and frequency within intervals. It does not show when each reading occurred.
2. The time plot shows readings that are fairly stable around 50 ppm for approximately the first 20 days, followed by a noisy upward drift through day 45. At about day 46, the readings suddenly return to values near 50 ppm.
3. Time order reveals whether the measurement process remains stable and when a change begins. Without it, the gradual loss of calibration and the later return to the target value could be mistaken for ordinary variation in one distribution.
4. The laboratory should investigate sensor or calibration drift, which could be related to sensor aging, contamination or changing environmental conditions. The sudden return near 50 ppm suggests that the sensor may have been recalibrated, cleaned, repaired or replaced around day 46; the maintenance log should be checked before drawing a conclusion.

Question 10

An economics report presents the illustrative data below. Average hourly wages and the consumer price level are both converted to indices with 2018 set equal to 100. Indexing allows their relative changes to be compared even though the original variables have different units.



1. Why is a time plot appropriate for these data? Identify the time variable and the two measured series.
2. Describe the trend in each series. Calculate the percentage change in average hourly wage and in the consumer price level from 2018 to 2026.
3. A headline claims, “**Workers are 40% better off than in 2018.**” Explain why the graph does not support this conclusion.
4. A simple purchasing-power index can be calculated as

$$\text{Purchasing-power index} = \frac{\text{wage index}}{\text{price index}} \times 100$$

Calculate this index for 2026 and interpret the result relative to 2018.

5. Identify one additional piece of information that would be needed before making a broad conclusion about whether workers are financially better off.

Solution

1. A time plot is appropriate because the observations are recorded across successive years and the goal is to study change over time. The time variable is year. The two measured series are the **average hourly wage** index and the **consumer price level** index.
2. Both series increase but the price level rises more rapidly during the middle years. The average wage changes from 20.00 dollars to 28.00 dollars, so its percentage increase is

$$\frac{28 - 20}{20} \times 100\% = 40\%$$

. The price index rises from 100 to 135, an increase of 35%.

3. The 40% figure is a nominal wage increase and does not account for the 35% increase in prices. Financial well-being depends partly on what wages can purchase, not only on the number of dollars earned. The graph also shows averages and therefore does not establish that every worker experienced the same wage change.
4. In 2026, the wage index is 140 and the price index is 135. Therefore

$$\frac{140}{135} \times 100 \approx 103.7$$

Under this simplified comparison, average purchasing power is about 3.7% higher than in 2018, not 40% higher.

5. Useful additional information could include the distribution of wage changes across workers, industries or regions; changes in hours worked; taxes and benefits; or whether the price index reflects the spending patterns of the population being discussed.

Question 11

A municipal public health office received restaurant related illness complaints from four districts during one year. The figures below are illustrative.

District	Number of restaurants	Complaints received
North	120	72
Central	800	160
South	240	96
West	90	45

A newspaper graph displays only the number of complaints and concludes that restaurants in the Central district present the greatest concern.

1. Why could comparing only the complaint counts be misleading?
2. Calculate the number of complaints per 100 restaurants for each district.
3. Which district has the largest complaint count? Which district has the highest complaint rate?
4. What graph would be appropriate for comparing complaint rates across districts? What additional information should accompany the graph?
5. Can these data establish that restaurants in the district with the highest complaint rate are less safe? Explain one limitation.

Solution

1. The districts contain very different numbers of restaurants. A district with more restaurants has more opportunities to generate complaints so a large count does not necessarily represent a large complaint rate.
2. The rates per 100 restaurants are

$$\frac{72}{120} \times 100 = 60$$

for North

$$\frac{160}{800} \times 100 = 20$$

for Central

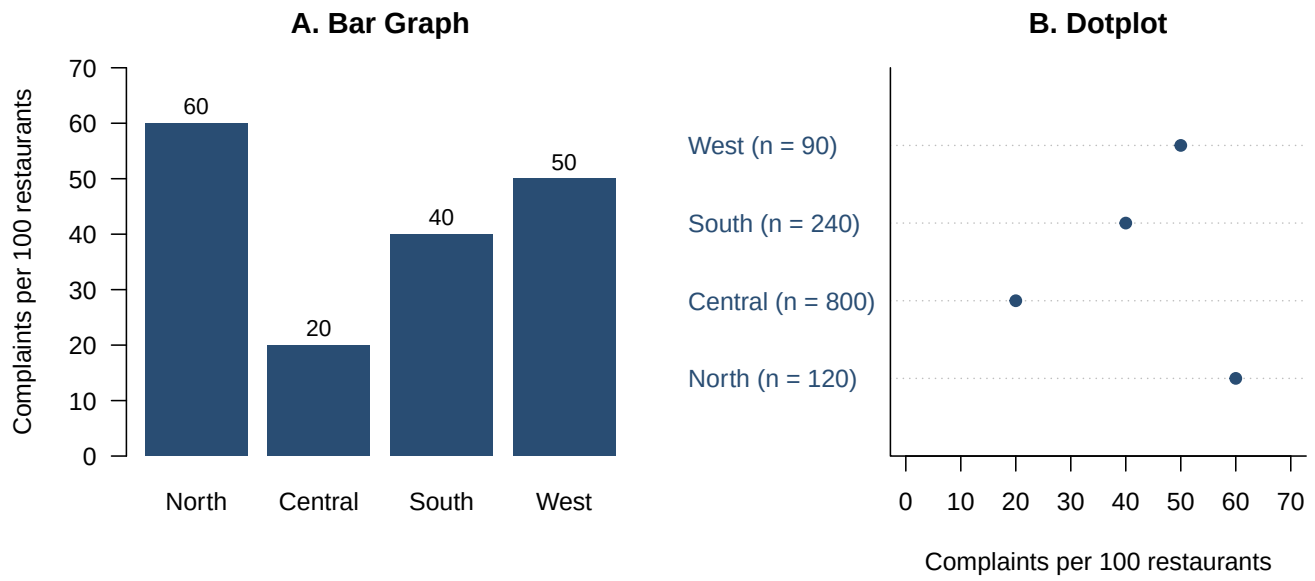
$$\frac{96}{240} \times 100 = 40$$

for South and

$$\frac{45}{90} \times 100 = 50$$

for West.

3. Central has the largest complaint count with 160 complaints. North has the highest complaint rate with 60 complaints per 100 restaurants.
4. A bar graph or dot plot of complaints per 100 restaurants would allow the rates to be compared. The number of restaurants and the time period should also be reported so that the denominators and scope are clear.



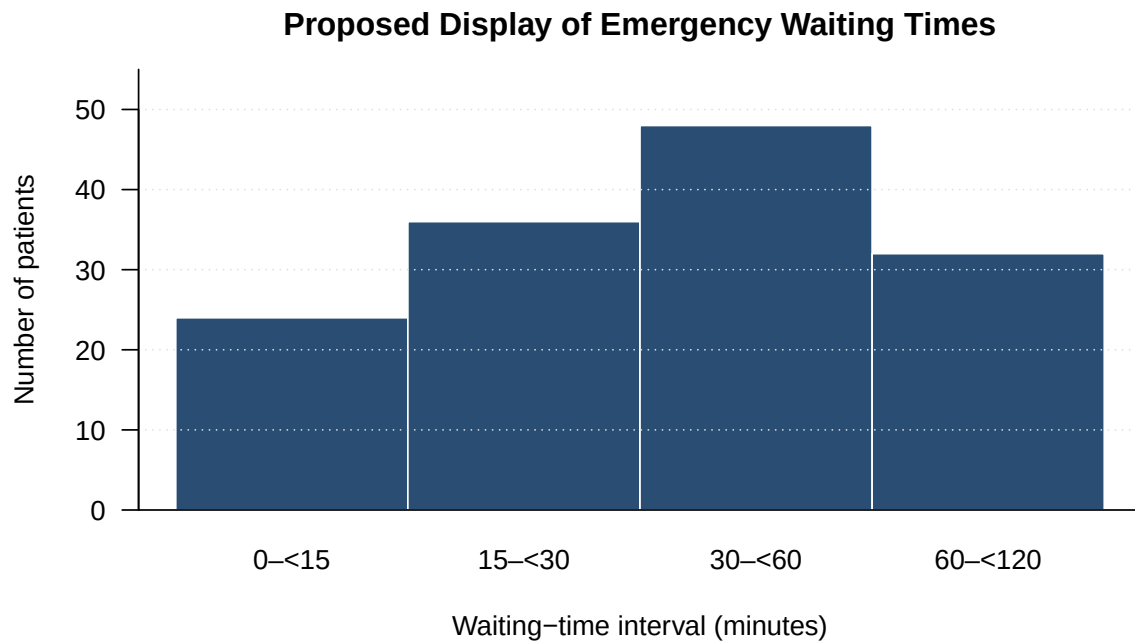
5. No. Complaint rates may be affected by differences in reporting behaviour, restaurant type, customer volume or inspection practices. The number of restaurants also does not measure how many customers each district served.

Question 12

An emergency department summarized the waiting times of 140 patients using intervals of different widths.

Waiting time (minutes)	Interval width	Number of patients
0 to under 15	15	24
15 to under 30	15	36
30 to under 60	30	48
60 to under 120	60	32

The proposed display gives every interval the same physical width.



1. Why is the proposed graph not a valid histogram?
2. Which interval contains the largest number of patients?
3. In a histogram with unequal interval widths, rectangle area should represent frequency. Calculate the frequency density for each interval using

$$\text{frequency density} = \frac{\text{frequency}}{\text{interval width}}.$$

4. Which interval would have the greatest height in the corrected histogram?
5. Describe how the horizontal and vertical axes should be constructed in the corrected display.

Solution

1. The numerical intervals do not have equal widths but the graph draws them as equally wide categories. This gives the 60 minute interval the same horizontal width as each 15 minute interval and distorts the distribution.
2. The interval from 30 to under 60 minutes contains the largest number of patients (48).

3. The frequency densities are

$$\frac{24}{15} = 1.6$$

$$\frac{36}{15} = 2.4$$

$$\frac{48}{30} = 1.6$$

and

$$\frac{32}{60} \approx 0.53$$

patients per minute, respectively.

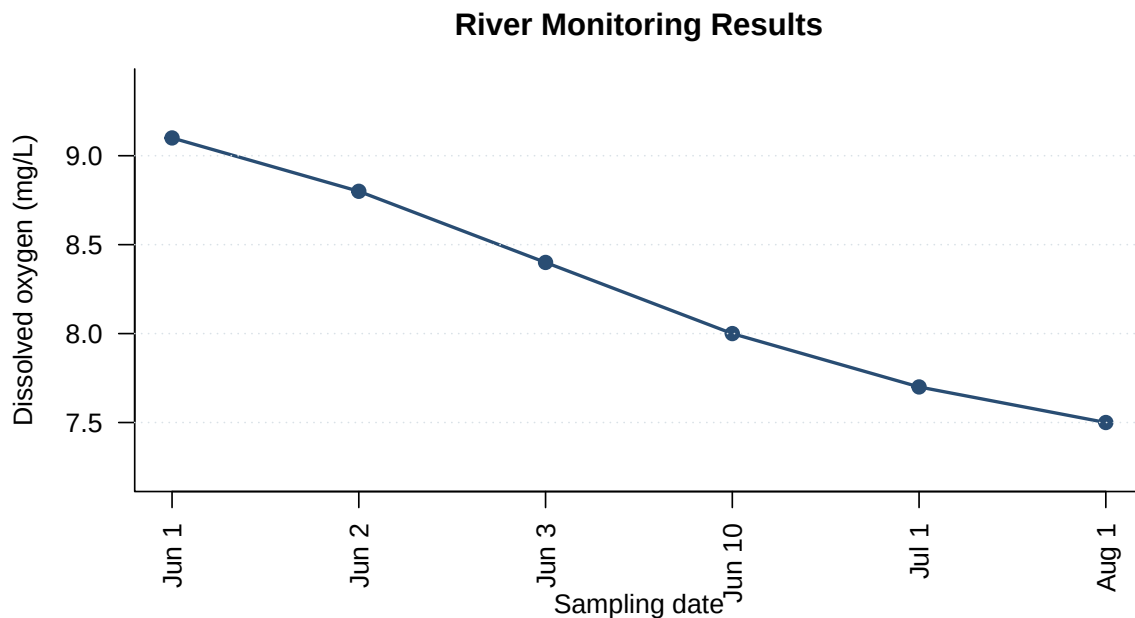
4. The interval from 15 to under 30 minutes would be tallest because it has the greatest frequency density, 2.4 patients per minute.
5. The horizontal axis should be a continuous waiting time scale from 0 to 120 minutes so the four rectangles have widths of 15, 15, 30 and 60 minutes. The vertical axis should show frequency density. Each rectangle's area will then correspond to the number of patients in its interval.

Question 13

An environmental team measured dissolved oxygen in a river on six dates. Lower dissolved oxygen concentrations can place aquatic organisms under stress. The measurements are illustrative.

Sampling date	Dissolved oxygen (mg/L)
June 1	9.1
June 2	8.8
June 3	8.4
June 10	8.0
July 1	7.7
August 1	7.5

The analyst produced the following time plot by placing consecutive observations at equal distances along the horizontal axis.



1. What impression does the equal spacing create about the rate of change over time?
2. Why is that impression misleading?
3. During which part of the study did dissolved oxygen decline most rapidly? Support your answer using the dates and measurements.
4. How should the horizontal axis be corrected?
5. Even after correcting the axis why should the connecting lines be interpreted cautiously?

Solution

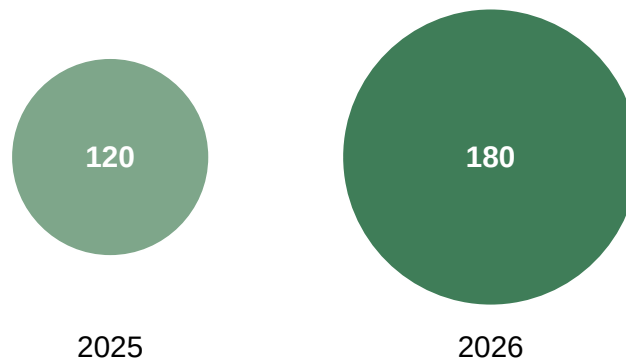
1. The equal spacing makes the measurements appear to decline at a fairly steady rate from June through August.
2. The actual gaps between measurements are 1 day, 1 day, 7 days, 21 days and 31 days. Treating these unequal gaps as equal visually stretches short periods and compresses long periods.
3. The most rapid observed decline occurred from June 1 to June 3. The concentration fell by 0.7 mg/L in two days. Later decreases occurred over much longer periods.

4. The measurements should be plotted against actual calendar dates on a continuous date axis. For example, the horizontal distance between July 1 and August 1 should be much larger than the distance between June 1 and June 2.
5. A line between two observations does not show what happened on the unmeasured days. Sudden changes or temporary recoveries could have occurred between sampling dates particularly across the longer gaps.

Question 14

A municipal report states that a recycling program diverted 120 tonnes of material from landfill in 2025 and 180 tonnes in 2026. An infographic represents each total with a circle whose **radius** is proportional to the number of tonnes.

Material Diverted from Landfill (tonnes)



1. By what factor did the reported tonnage increase from 2025 to 2026? What was the percentage increase?
2. Because viewers perceive the **area** of each circle, calculate the ratio of the displayed circle areas. Recall that the area of a circle is $A = \pi r^2$.
3. How does the visual comparison exaggerate the actual change?
4. Describe two fairer ways to display the two values.
5. If circles must be used, by what factor should the 2026 radius increase so that circle area, rather than radius, is proportional to tonnage?

Solution

1. The tonnage increased by a factor of

$$\frac{180}{120} = 1.5$$

. This represents a

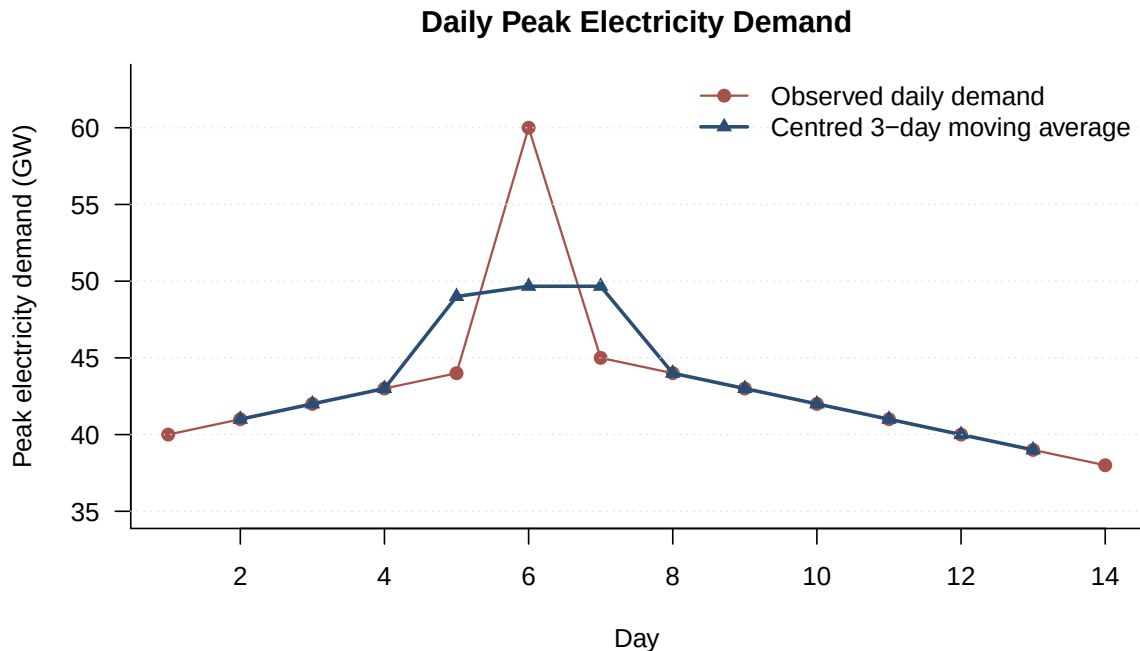
$$\frac{180 - 120}{120} \times 100\% = 50\%$$

increase.

2. The radius ratio is 1.5 so the area ratio is $1.5^2 = 2.25$. The 2026 circle therefore appears to have 2.25 times the area of the 2025 circle.
3. The data increased by 50% but the displayed area increased by 125%. The graphic therefore makes the increase appear substantially larger than it was.
4. A bar graph beginning at zero would provide a fair comparison. A labelled dot plot on a common numerical scale would also work. A simple table with both values is sufficient when only two totals are compared.
5. The radius should increase by $\sqrt{1.5} \approx 1.225$, or about 22.5%, because squaring this radius factor produces the required area factor of 1.5.

Question 15

An electricity grid operator recorded the daily peak demand during a two-week heat event. A report displays both the observed daily values and a centred three day moving average. The data are illustrative.



1. Why is a time plot appropriate for these data?
2. What is the largest observed daily demand and on which day did it occur?
3. Calculate the centred three day moving average for day 6. How does it compare with the observed value on that day?
4. A summary based only on the moving average line states “**Demand remained below 50 GW throughout the event**”. Is this statement accurate? Explain.
5. Which series is more important for short-term capacity planning, and which is more useful for describing the underlying trend? What information about the moving average should appear in the graph or caption?

Solution

1. A time plot preserves the order of the daily observations and reveals changes, unusual peaks and short-term patterns across the event.
2. The largest observed daily demand is 60 GW on day 6.
3. The centred moving average for day 6 is

$$\frac{44 + 60 + 45}{3} \approx 49.7$$

GW. Averaging the spike with the adjacent days reduces its displayed height by approximately 10.3 GW.

4. No. The observed demand reached 60 GW on day 6. The statement describes the smoothed line but incorrectly presents it as a statement about the actual daily peaks.
5. The observed series is essential for capacity planning because the grid must be able to meet short lived peaks. The moving average is useful for describing the broader trend. The graph or caption should identify the smoothing method, the three day window and whether the average is centred or trailing. Displaying the raw observations alongside the smoothed series prevents the peaks from being hidden.